

Table 3-4. Coverage of Student Outcomes 1-7 by CHE 494 Selected Topic Courses.

Course	(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) An ability to communicate effectively with a range of audiences.	(4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CHE 494-A	1	3	4	4	4	1	4
CHE 494-B	2	1	3	4	4	4	4
CHE 494-C	2	1	3	4	4	4	4
CHE 494-D	4	3	4	1	1	3	4
CHE 494-E	4	3	2	3	2	4	3
CHE 494-F	4	4	4	1	3	3	4
CHE 494-G	3		3				2

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

CHE 494 A Entrepreneurship in Chemical Engineering

CHE 494 B Graphene and 2D Materials

CHE 494 C Materials for Energy

CHE 494 D Molecular and Macromolecular Engineering

CHE 494 E Nano and Micro Engineered Membranes

CHE 494 F Nanotechnology and Bio-Nanotechnology

CHE 494 G Polymer Processing and Technology